

DEEP LEARNING FUNDAMENTALS

Training a Transformer

Forward Pass, Loss Function, and
the Mechanics of Backpropagation



HOW DOES A MODEL LEARN?



1. Forward Pass

The model takes input data, processes it through neural layers, and makes a mathematical "guess" or prediction.



2. Compute Loss

We compare the guess with the actual truth. The difference is measured by a scalar value called



3. Backpropagation

We trace the error back through every parameter to see how much each weight contributed to the mistake.



4. Weight Update

Using $\frac{\partial L}{\partial w}$, we nudge billions of weights in the direction that minimizes the loss.

THE FORWARD PASS

During training, data flows from the **input embeddings** through a stack of layers.

 **Multi-Head Attention:**

Mixing context between tokens.

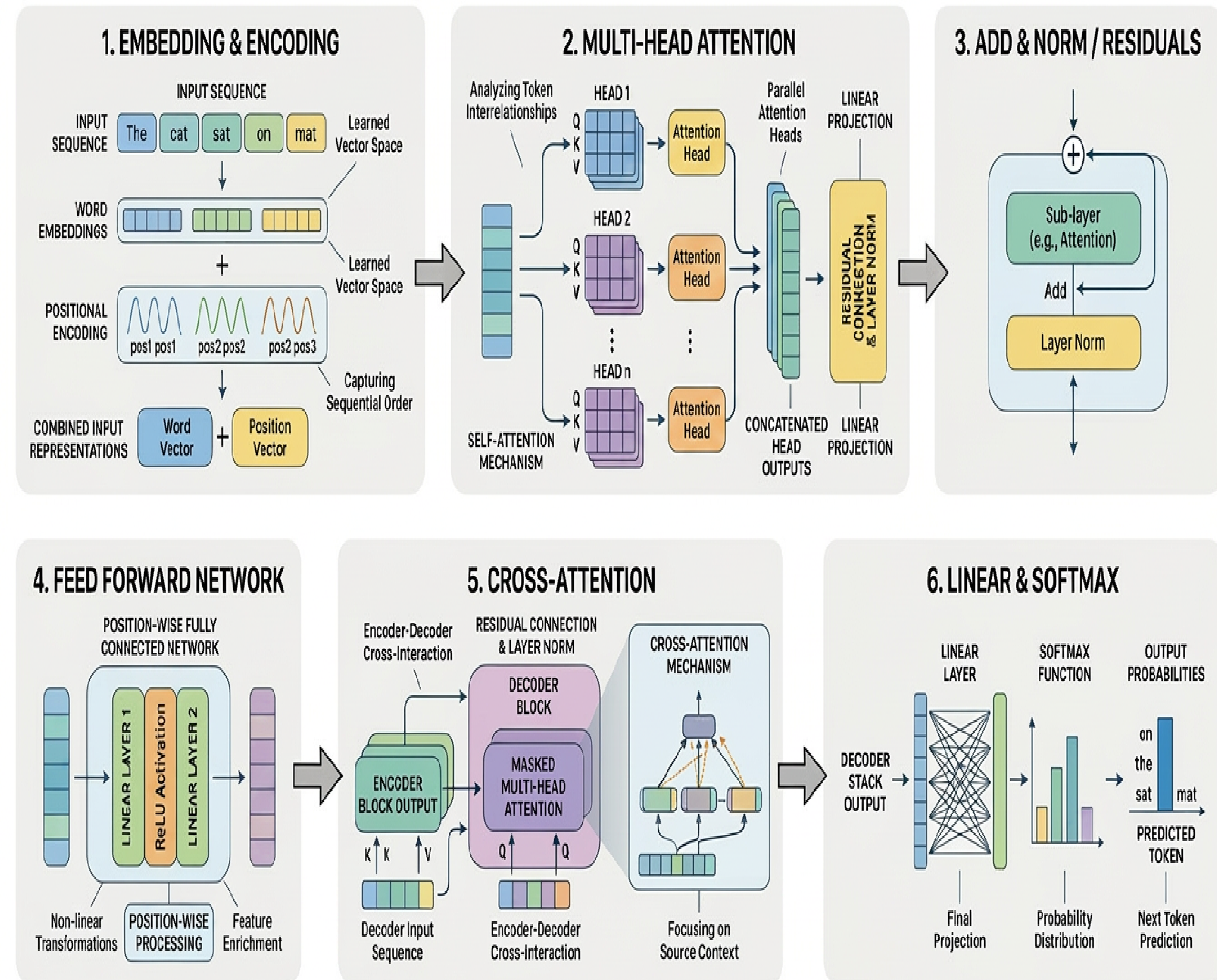
 **Feed Forward Networks:**

Transforming representations.

 **Logits Generation:**

Producing raw scores for the entire vocabulary.

TRANSFORMER MODEL ARCHITECTURE



CONVERTING RAW SCORES

The Final Layer

At the end of the transformer stack, we have a **Linear Projection**. It maps the high-dimensional hidden state to a vector of size V (Vocabulary size).

$$z_i = W_{\text{out}} h + b$$

What are Logits?

Logits (z_i) are raw, unbounded numbers. Higher values represent tokens the model believes are more likely to appear next.

Example:

"Paris" -> 12.5

"Tree" -> -2.1

"Cat" -> 1.4

SOFTMAX: THE PROBABILIZER

We cannot calculate loss directly from raw logits. We need to convert them into a **Probability Distribution** where all values sum to 1.0.

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^V e^{z_j}}$$

100%

Sum of all outputs

Exponential scaling ensures that even small differences in logits lead to high confidence in specific tokens, making the model more "decisive."

MEASURING THE ERROR

Training requires a "Scoreboard." We use **Cross-Entropy Loss** to penalize the model when its predicted distribution differs from the target (one-hot) distribution.

Token	Target (Truth)	Prediction (Guess)	Result
The	1.0	0.85	Good Match (Low Loss)
And	0.0	0.10	Slight Error
Tree	0.0	0.05	Minimal Error

CROSS-ENTROPY FORMULA

$$L = - \sum_{i=1}^V y_i \log (p_i)$$

- ✓ y_i : The ground truth (usually 1 for correct word, 0 otherwise).
- 📈 p_i : The model's probability for that word.
- i **Intuition:** If $p_i = 1$ for the correct word, $\log (1) = 0$, **Loss** = 0.

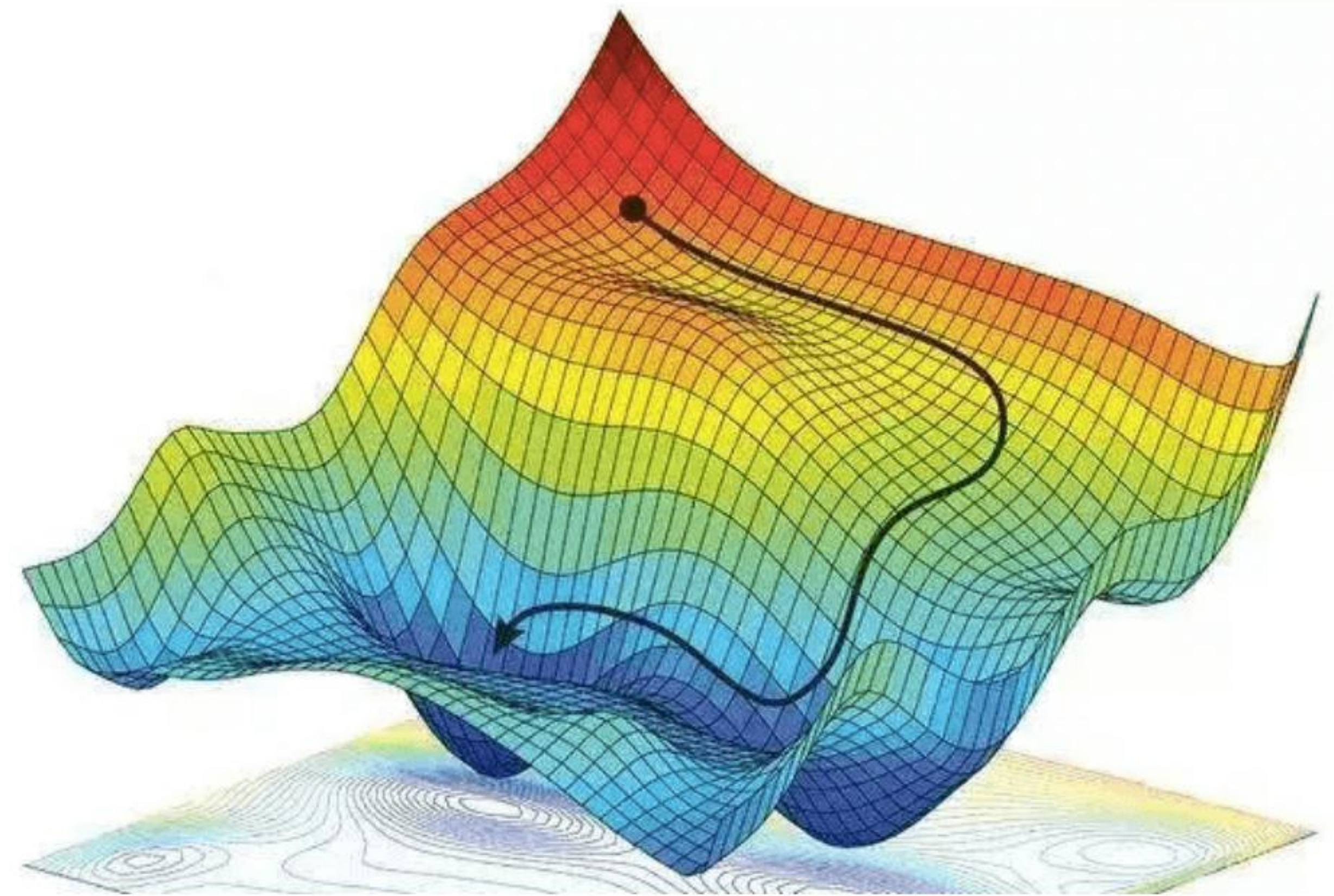
THE OPTIMIZATION VALLEY

Think of training as finding the lowest point in a massive mountain range. Each weight is a coordinate, and the **Loss** is the altitude.

Goal: Find the global minimum (Lowest Loss).

Method: Step downward in the steepest direction.

Gradient: The vector pointing "uphill." We move in the opposite direction.



BACKPROPAGATION SEQUENCE



1. Loss Gradient

Calculate $\frac{\partial L}{\partial p}$



2. Chain Rule

Propagate error backward through softmax to the last layer.



3. Layer Error

Calculate gradients for every weight in every transformer block.



4. Accumulate

Combine gradients across the entire batch of text.

"How much did this specific weight contribute to the final error?"

UPDATING THE BILLION PARAMETERS

Once we have the gradient (∇L), we update the weights (θ) using the **Learning Rate** (η).

$$\theta_{t+1} = \theta_t - \eta \nabla L$$

By repeating this billions of times, the model "memorizes" patterns and "understands" relationships in language.

Optimization Stats

Parameters: 175B+

Steps: 1M+

Precision: BF16

CONVERGENCE AT SCALE



Loss typically follows a "Power Law" — initial rapid gains followed by long-term gradual optimization.

Conclusion

Learning is the iterative process of minimizing error through mathematical optimization.

Summary

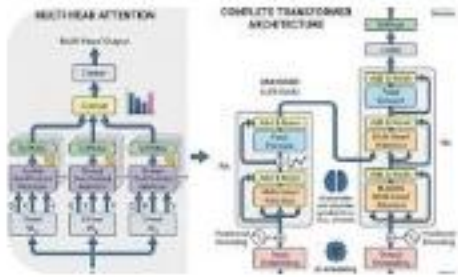
- Forward pass makes predictions.
- Loss quantifies error.
- Backprop assigns blame.
- Optimizers update weights.

Contact Info

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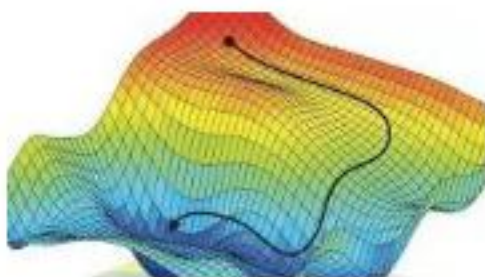
GitHub: [/transformer-training](#)

IMAGE SOURCES



<https://scispace.com/ai-agent-seo/page-thumbnails/build-transformer-model-arc-gmduxirb.png>

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